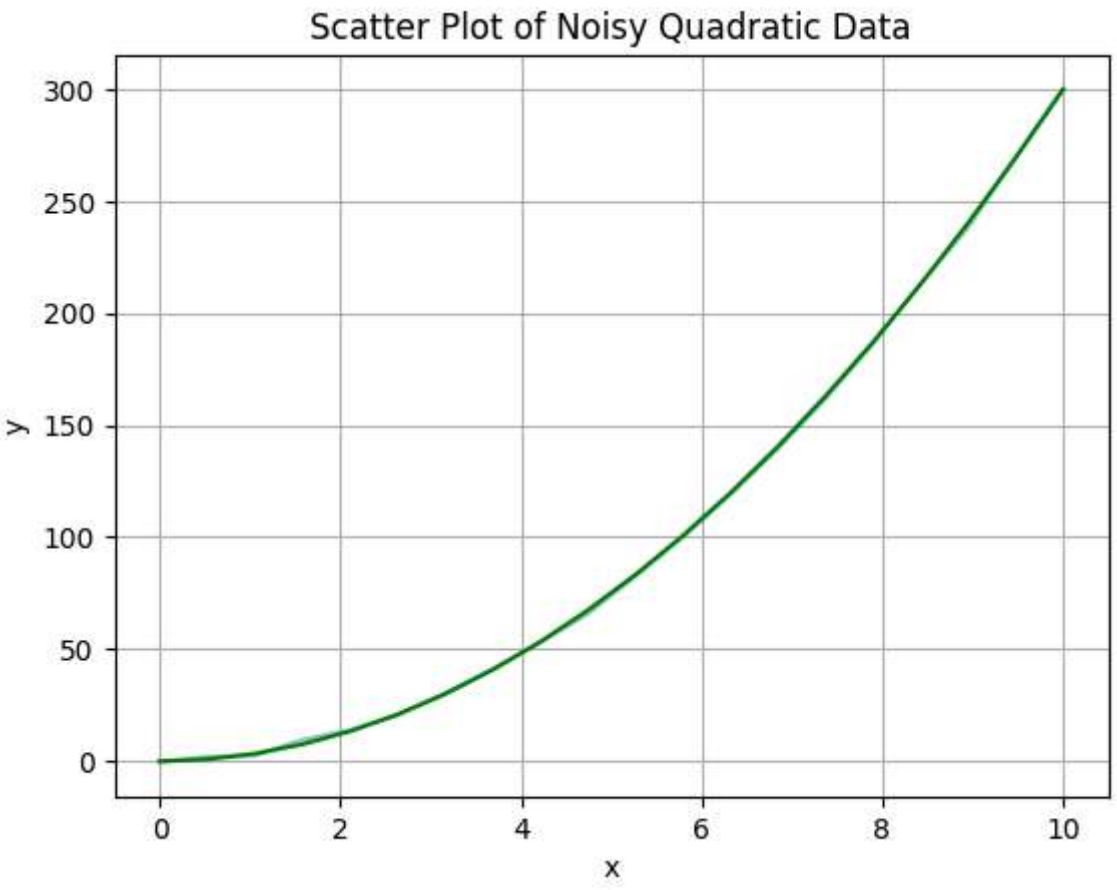
Fit y = 3x^2 + 5 + N(0,1) with polynomial functions

In [6]:
```
import numpy as np
import matplotlib.pyplot as plt
```

Original curve with noise

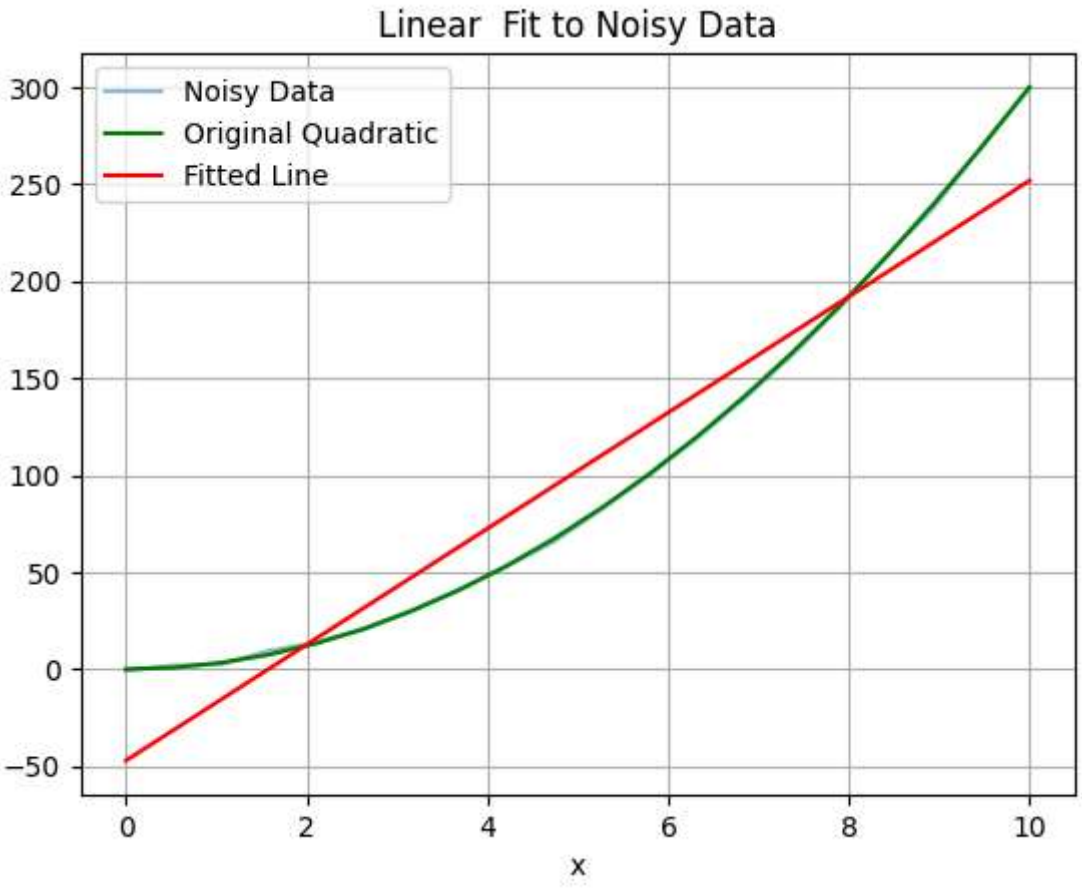
In [7]:
```
x = np.linspace(0, 10, 20)
y = 3 * x**2 + np.random.normal(0, 1, size=x.shape)
y_og = 3 * x**2
plt.plot(x, y, alpha=0.5)
plt.plot(x, y_og, color='green', label='Original Quadratic')
plt.title('Scatter Plot of Noisy Quadratic Data')
plt.xlabel('x')
plt.ylabel('y')
plt.grid()
plt.show()
```



Fittnig 1-D line

In [8]:
```
X = np.column_stack((np.ones_like(x), x))
theta = np.linalg.inv(X.T @ X) @ X.T @ y
y_pred = X @ theta
print("Coefficients:", theta)
plt.title('Linear  Fit to Noisy Data')
plt.xlabel('x')
plt.plot(x, y, alpha=0.5, label='Noisy Data')
plt.plot(x, y_og, color='green', label='Original Quadratic')
plt.plot(x, y_pred, color='red', label='Fitted Line')
plt.legend()
plt.grid()
plt.show()
```

```
Coefficients: [-47.05702537  29.88043086]
```

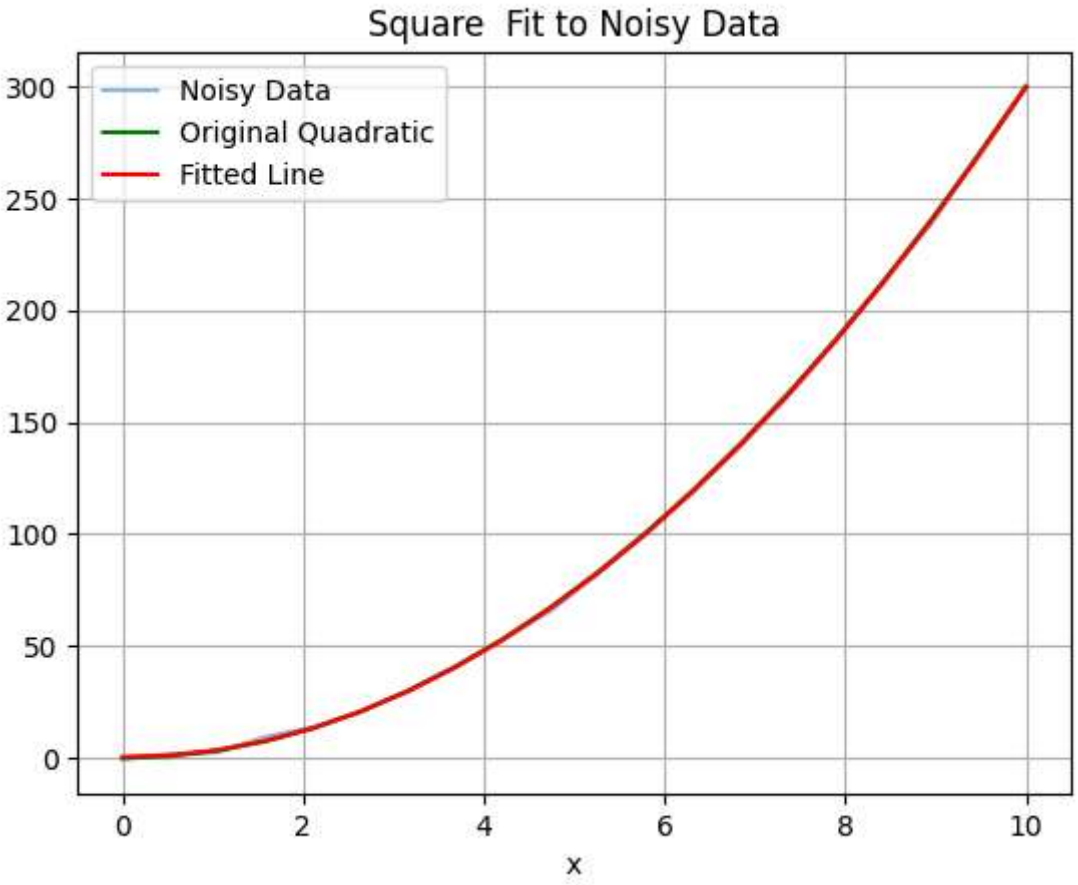


Fitting square curve

In [9]:
```
X = np.column_stack((np.ones_like(x),x, x**2))
theta = np.linalg.inv(X.T @ X) @ X.T @ y
y_pred = X @ theta
print("Coefficients:", theta)
plt.title('Square  Fit to Noisy Data')
plt.xlabel('x')
plt.plot(x, y, alpha=0.5, label='Noisy Data')
plt.plot(x, y_og, color='green', label='Original Quadratic')
plt.plot(x, y_pred, color='red', label='Fitted Line')
plt.legend()
```

```
plt.grid()
plt.show()
```

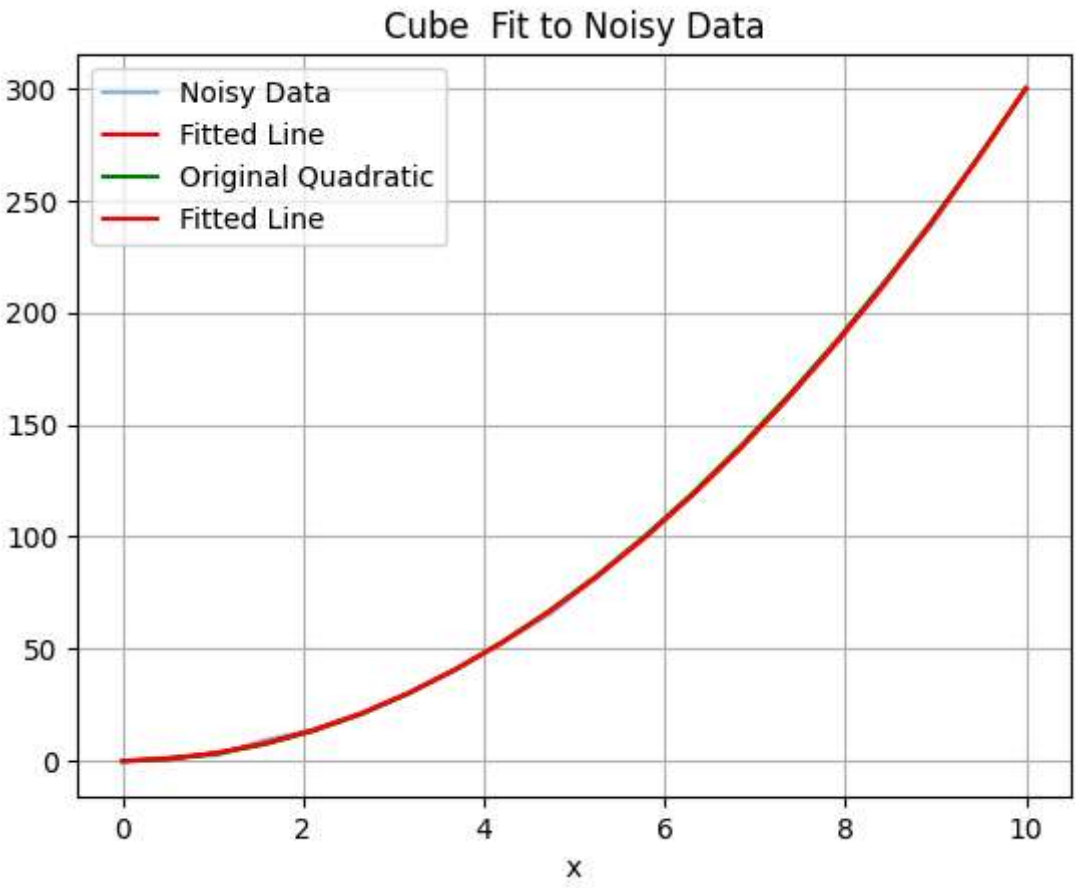
```
Coefficients: [ 0.57827096 -0.28859014  3.0169021 ]
```



Fitting cubinc curve

In [10]:
```
X = np.column_stack([np.ones_like(x), x, x**2, x**3])
theta = np.linalg.inv(X.T @ X) @ X.T @ y
y_pred = X @ theta
plt.plot(x, y, alpha=0.5, label='Noisy Data')
plt.plot(x, y_pred, color='red', label='Fitted Line')
print("Coefficients:", theta)
plt.title('Cube  Fit to Noisy Data')
plt.xlabel('x')
plt.plot(x, y_og, color='green', label='Original Quadratic')
plt.plot(x, y_pred, color='red', label='Fitted Line')
plt.legend()
plt.grid()
plt.show()
```

```
Coefficients: [0.00383486 0.5039815  2.81359677 0.01355369]
```


Cube Fit to Noisy Data
Noisy Data
Fitted Line
Original Quadratic
Fitted Line
300
250
200
150
100
50
0
0
2
4
6
8
10
x

In [ ]: